PROJECT 18 · SEMESTER 5 APPLIED ML & DEEP LEARNING

Surface Water Quality Prediction

Estimating the CCME Water Quality Index with gradient-tree ensembles and aquatic-chemistry features — plus an honest audit of what the scores mean



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2.82 M observations · 5 countries · 1940–2023

THE PROBLEM

Lab assays are slow and sparse; an index estimator could triage faster

Grab sample

Field visit, discrete in space and time

→

Lab assay

BOD5 needs 5 days (120 h) incubation

→

Index

Composite score computed after results arrive

Project objective

Learn a fast surrogate: eight measured parameters (pH, temperature, DO, BOD, ammonia, orthophosphate, nitrogen, nitrate) plus engineered features → CCME WQI score.

Scope of claims

- Demonstrated: offline regression on a historical dataset (Random Forest, XGBoost)
- Not demonstrated: real-time sensing, field deployment, replacing laboratory testing, or any neural network (planned Phase 2)

OBJECTIVE AND TARGET

The target is the CCME Water Quality Index: 100 minus a distance from ideal

$$\text{CCME WQI} = 100 - \sqrt{(F_1^2 + F_2^2 + F_3^2)} \,/\, 1.732$$

F_1 Scope

% of variables that fail their guideline

F_2 Frequency

% of individual tests that fail

F_3 Amplitude

Size of excursions: nse / (0.01·nse + 0.01)

Five quality tiers (share of dataset rows)

Poor 0–44	Marginal 45–64	Fair 65–79	Good 80–94	Excellent 95–100
4.5%	11.6%	11.2%	36.6%	36.1%

Index in the data: mean ≈ 85, median 90.6, bounded 0–100, strongly left-skewed. Shares from repo graph 11 (EDA).

DATASET SCALE AND COMPOSITION

2.82 million observations, but 75% of them come from England

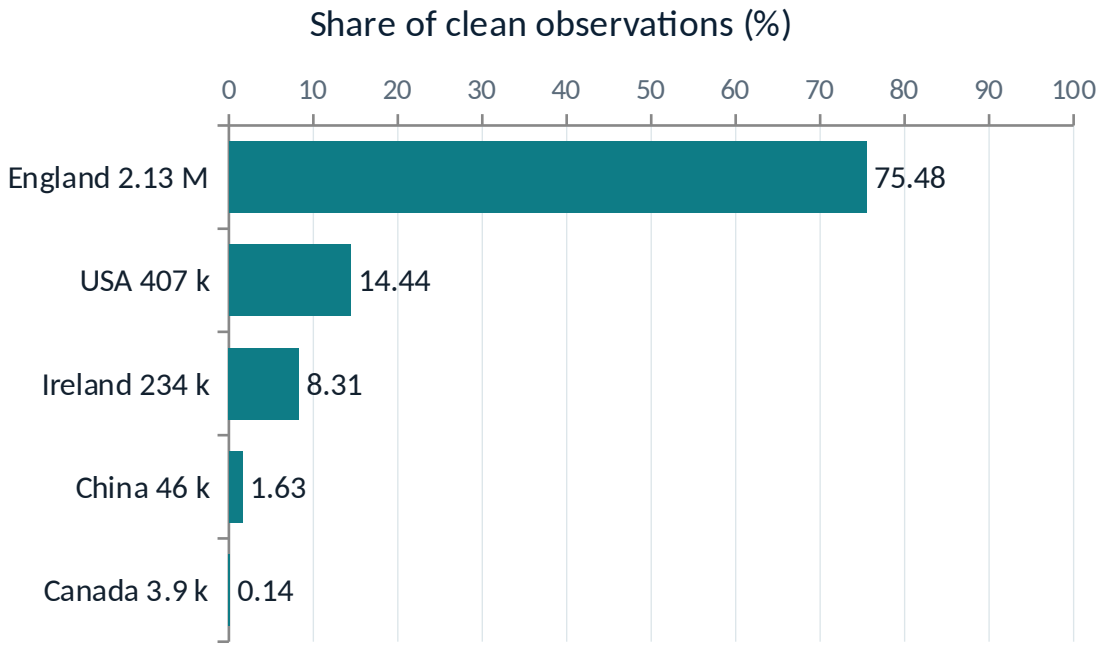
2,827,977
raw records

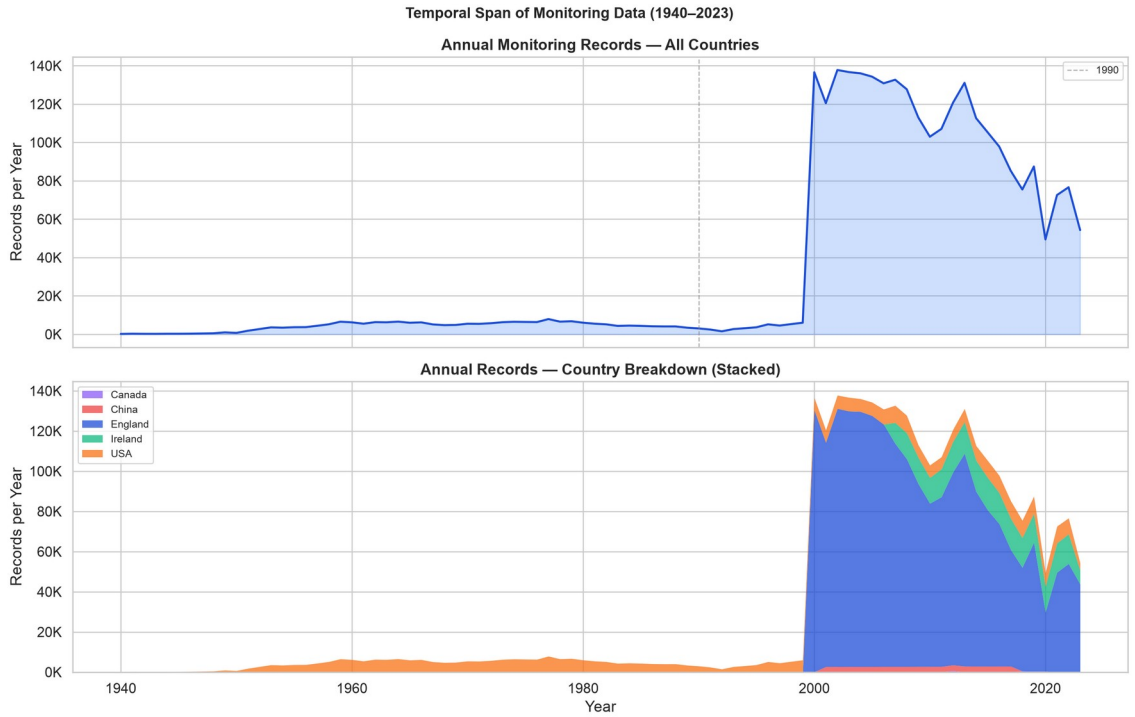
2,819,617
clean records

5
countries

1940–2023
83 years

38
model input features





Records per year (repo graph 03): coverage is sparse before ≈2000

DATA AUDIT AND PHYSICAL-BOUND SCREENING

8,360 impossible readings removed; zero nulls in the file, but one completeness question



Parameter	Valid range	Violations
pH	3.0 – 12.0	1,581
Temperature (°C)	0 – 45	6,850
Dissolved O_2 (mg/L)	0 – 25	0
Ammonia / BOD (mg/L)	≥ 0	2 / 2
Orthophosphate (mg/L)	≥ 0	10
Nitrogen / nitrate	≥ 0	0 / 0

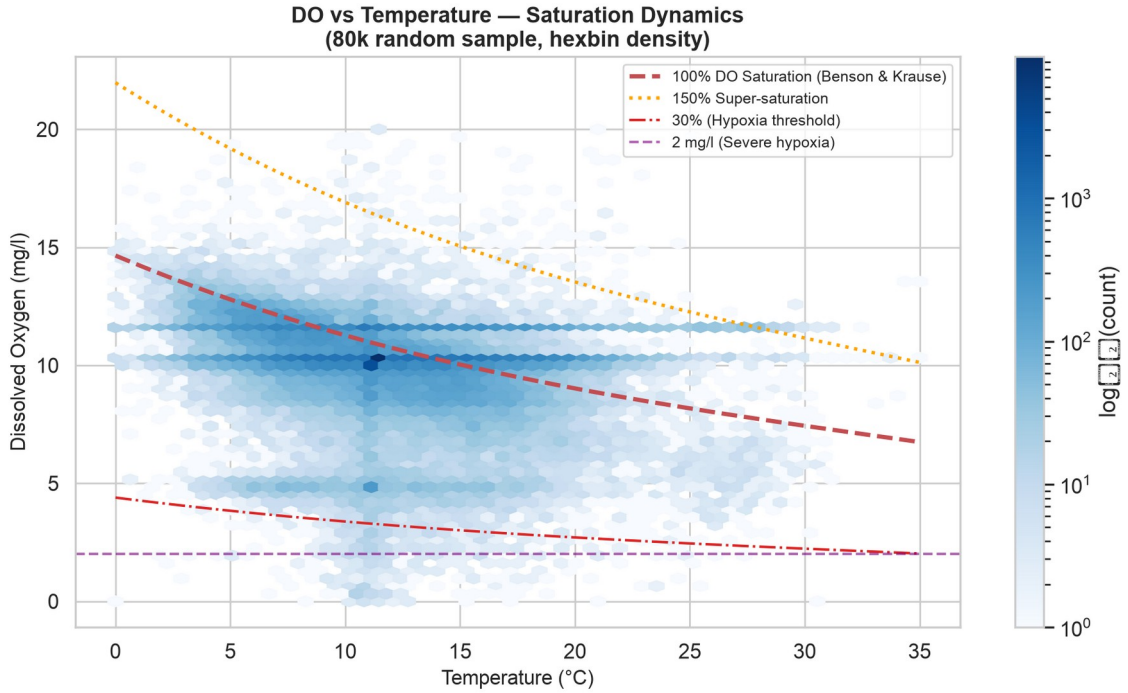
Counts overlap; 8,360 is the unique union of violating rows (executed EDA notebook).

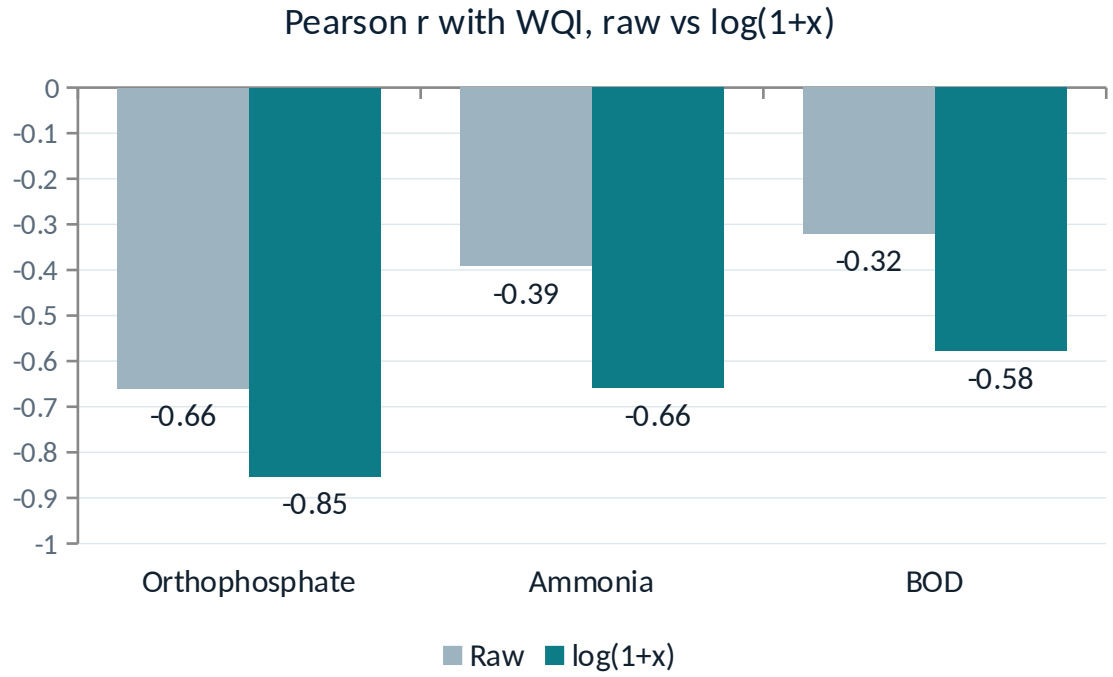
Missing vs implausible. No imputation was applied in our pipeline; screening removes sensor artefacts (negative or out-of-range values), not gaps.

Qualification "0 nulls" describes the delivered file. Values cluster (BOD median 2.70, Q3 2.83) and repo graph 17 (earlier version) showed 26–34% missing, so upstream filling is possible and unverified.

EXPLORATORY FINDINGS

Oxygen follows temperature; nutrients are heavily skewed, and log(1+x) straightens them





DO vs temperature (80k sample) with Benson–Krause saturation curves. Source: repository graphs/ folder (graph 07).

Qualification A better Pearson r is not better prediction. Orthophosphate's Spearman ρ is −0.917 both raw and log(1+x): monotone transforms leave rank correlation, and tree splits, unchanged.

DOMAIN-INFORMED FEATURE ENGINEERING

Eight measurements become 38 features, grouped by the chemistry they encode

8

Base sensors

pH, temperature, DO, BOD, ammonia, orthophosphate, nitrogen, nitrate

7

Aquatic stoichiometry

Benson–Krause DO deficit and % saturation, Redfield-style N:P, Emerson un-ionised NH_3, BOD:DO, pH deviation, nitrate share

7

log(1+x) variants

Compress nutrient skew; help linear models, not trees

6

Station memory

Shift-1 lags and rolling-3 means within each Area, sorted by date; first record filled with global median

7

Temporal

year, month, quarter, sin/cos of month and day-of-year

2

Context

country code, waterbody-type code

1

exceedance_counter

Count of six guideline checks computed from input columns only

Total: 8+7+7+6+7+2+1 = 38 features

ρ(hist_ortho_lag1, WQI) = −0.78 (Spearman, full data)

PIPELINE ARCHITECTURE AND TESTING

Features are built before the split, so station history can cross partitions

1 Load cleaned data
2,819,617 rows, 0 nulls

2 38 features
lags built on full data

3 Target check
targets not in features

4 Stratified split
random, by WQI tier

5 Train RF + XGB
XGB early-stops on val

6 Evaluate
validation only

7 Importance
built-in + SHAP (250 rows)

8 Test set
held out, not yet scored

Navy = before split · Amber = leakage-sensitive · Teal = after split · Grey = not yet done

Automated tests (pytest)

12 test functions: features 4 · leakage 2 · metrics 3 · models 3. Reported 100% pass in 1.65 s (not re-run here).

They check: bound filtering, log1p, shift-1 lags per station, targets absent from X and from the 38-feature list, metric maths, model loading.
They do not show: that scores generalise to new stations or dates.

MODELING STRATEGY AND EVALUATION DESIGN

Three regressors on one random stratified split, with the settings taken from the training notebook



Random, stratified by WQI tier, seed 42 · no station or time hold-out · reported metrics are on validation; the test set has not been scored

Mean baseline

Predicts the training mean (85.05). The floor any model must beat.

No fitting beyond the mean

Random Forest

30 trees, max depth 14, min leaf 5, all 38 features considered per split, seed 42.

Notebook = saved model

XGBoost

≤150 rounds, depth 6, η 0.08, subsample 0.8, colsample 0.8; early stopping (20) on validation; best round 149.

Early stopping uses the validation set

Neural network

Not implemented in this repo version. Planned for Phase 2 (multi-task MLP).

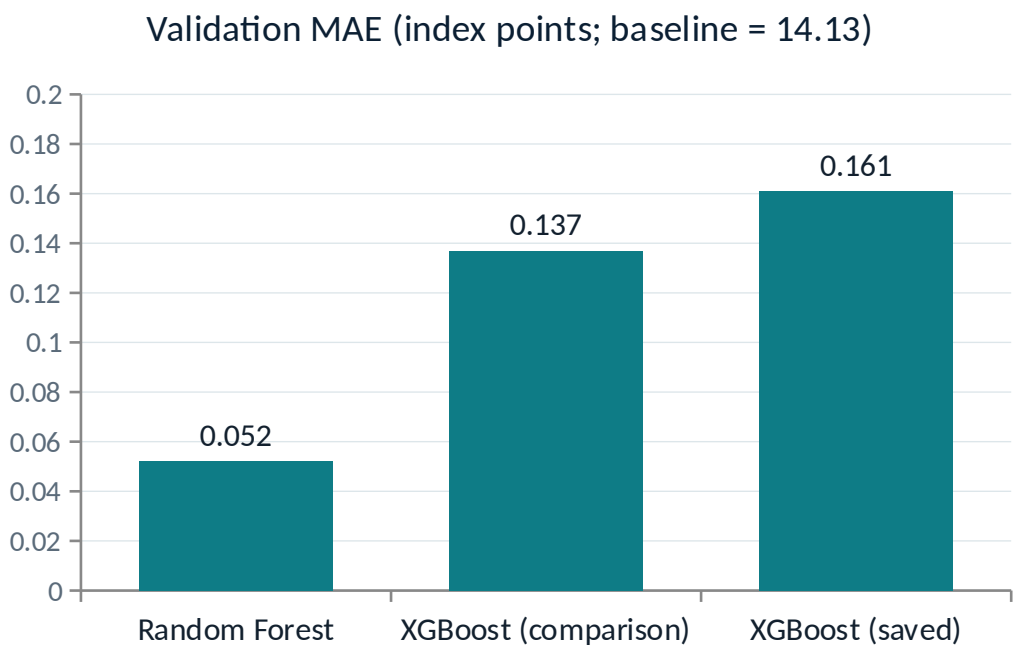
No results claimed

MODEL BENCHMARKING

Tree ensembles reproduce the index almost exactly on validation rows

Model	MAE	RMSE	R^2
Mean baseline	14.130	17.656	0.0000
Random Forest	**0.052**	**0.208**	**0.9999**
XGBoost (comparison file)	0.137	0.374	0.9996
XGBoost (saved model run)†	0.161	0.437	0.9994

† Training notebook and the saved XGBoost model's recorded validation RMSE (0.4372). The saved predictions file behind the comparison table disagrees; provenance unresolved. All values: validation set, 422,943 rows.



Qualification These scores measure how well each model reconstructs the CCME index from same-time measurements on randomly held-out rows (the repo's own notebook says so). They do not show forecasting skill. Latency omitted: not measured in the repo.

RELIABILITY AND VALIDITY

Why $R^2 \approx 0.9999$ needs careful reading: what the code shows

1 Target is a formula of the inputs

CCME_Values ships with the data and is a composite of the same eight measurements. High R^2 means the model learned the formula. The repo's evaluation notebook says exactly this.

Confirmed in repo

2 exceedance_counter re-checked

Built only from six input columns with its own thresholds; not a target column, not F_1–F_3 (not in the data). Legitimate but redundant. Ablation not yet run.

Re-assessed: no data-flow leak

3 Random split, early stopping

Stratified random split; per-Area lags built on full data, so neighbours cross partitions; XGBoost stops on validation; test set unscored.

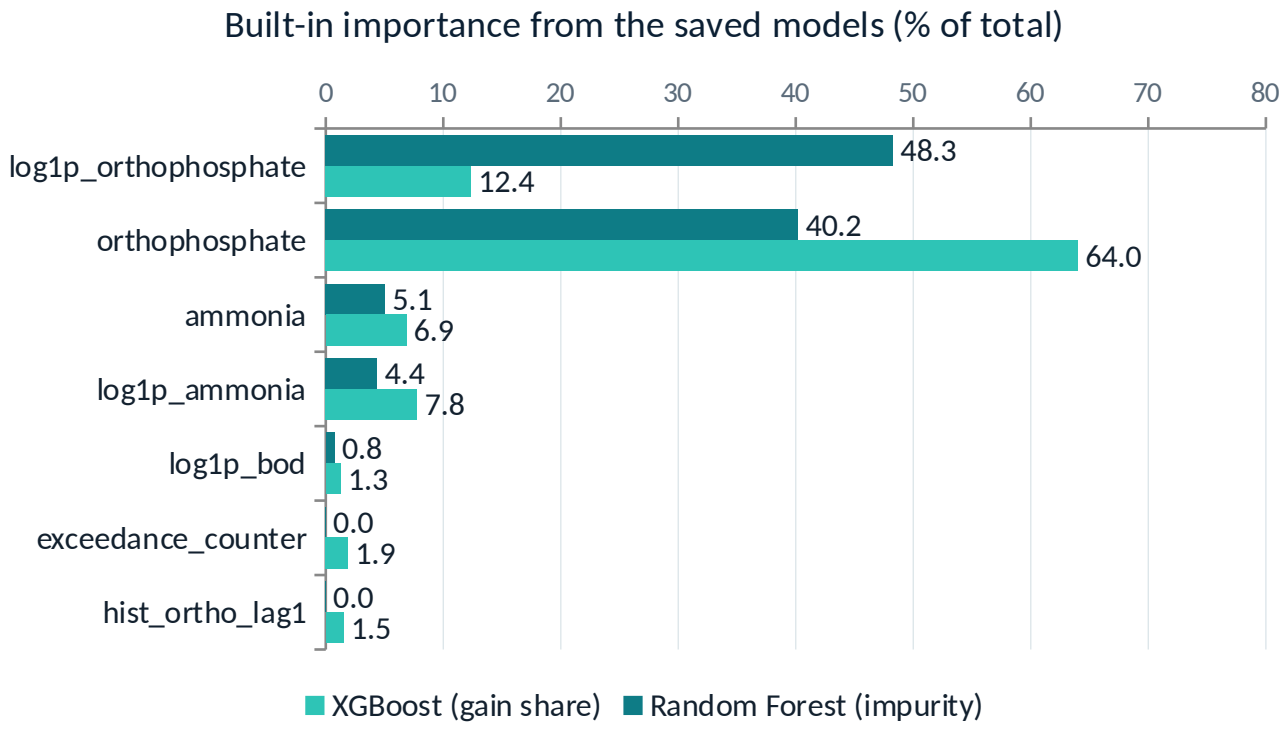
Confirmed in code

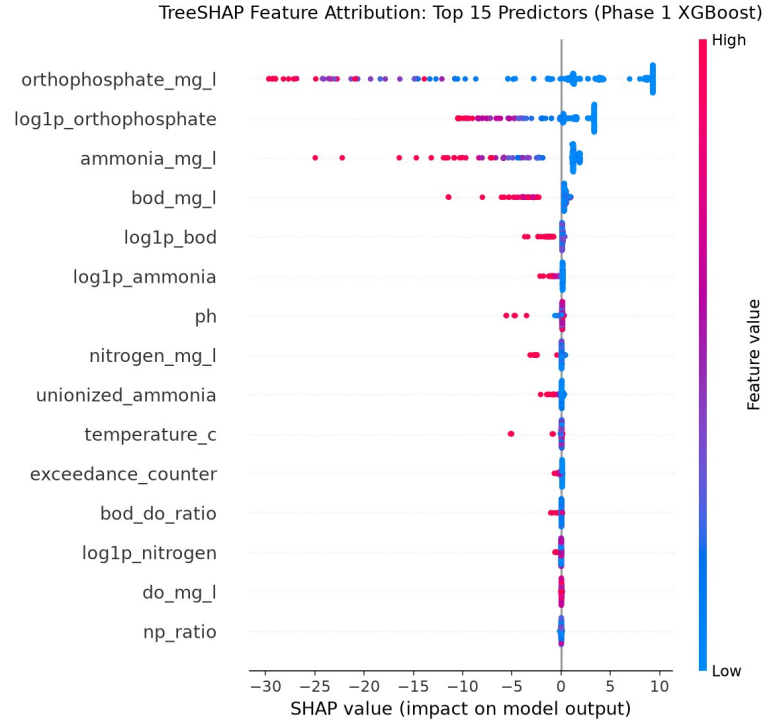
Can claim
A fast, accurate surrogate of the CCME computation on this dataset (validation rows).

Cannot claim
Performance on unseen stations or future dates; independent water-quality forecasting.

WHAT DRIVES THE PREDICTIONS

Orthophosphate dominates the models — a property of the index and data, not a causal finding





TreeSHAP beeswarm, XGBoost, 250 validation rows (repo graph 16)

Qualification Importance describes the model, not the river. Orthophosphate + log variant: 88.5% of RF impurity, 76.4% of XGBoost gain; not “88% of split decisions”. The station lag ranks 6th in XGBoost (1.5%).

Phase 1 delivers a tested tree baseline; stricter validation and deep learning come next

Confirmed from code and outputs

- Country imbalance: England ≈75% of rows
- Random split; no station or time hold-out
- Validation metrics only; test set unscored
- XGBoost validation numbers differ between artifacts

Open / not verified

- Whether the 0-null file reflects upstream filling (clustered values; graph 17)
- Effect of dropping exceedance_counter
- Duplicates across partitions
- Inference latency (not measured)

Planned (not done)

- Station-held-out and time-forward splits
- Score the held-out test set
- Re-run notebooks 03–04 to align artifacts
- Phase 2: multi-task neural network

Deep learning: no neural-network code or results exist in this repo version, so none are claimed. Phase 2 is planned.

A reproducible tree baseline — with a clear route to a stricter evaluation

Built: audited 2.82 M-row dataset, 38 chemistry-driven features, RF and XGBoost benchmarks, SHAP, 12 tests.

Strongest finding: trees reconstruct the CCME index with validation MAE 0.05 (RF) to 0.14–0.16 (XGBoost).

Key caveat: the target is a formula of the inputs and the split is random; next, station-held-out and time-forward evaluation.

Team and roles

Mayank Sharma – EDA & repository management lead
Aman Soni – model training lead (baseline, RF, XGBoost)
Aaryan Gera – data preprocessing
Chaitanya Kumar – feature engineering & pipeline
Ankit Kumar Pandey – model evaluation & validation
Pranav Sehgal – research, documentation & presentation

Key references

Breiman (2001) Random Forests. Machine Learning 45(1)

Goodfellow, Bengio & Courville (2016) Deep Learning. MIT Press (feed-forward networks)

Chen & Guestrin (2016) XGBoost. KDD

Lundberg & Lee (2017) SHAP. NeurIPS

CCME (2001) Water Quality Index 1.0 Technical Report

Benson & Krause (1984) Limnol. Oceanogr. 29(3)

Redfield (1958) Am. Scientist 46(3)

Emerson et al. (1975) J. Fish. Res. Board Can. 32(12)

Data: agencies listed in the repo's DataCollectionSources sheet (data.gov.uk, USGS/EPA, EPA Ireland, MEE China, ECCC). Code: github.com/MAYANKSHARMA01010/Semester_5_AML_DL